

Quantization and Distillation

What is Quantization and Distillation?

Distillation: It is the process of transferring the knowledge acquired by a large ensemble of models or an extremely large model to a smaller model.

What it means is that we should create two models. One is a training model and the other should be an inference model. The training model should be large to extract features from the training data and can have latency. After this we will create a smaller model and then pass the knowledge gathered from the larger model to the smaller model which is more suitable for deployment. This smaller model will not only learn the hard labels from the dataset but also the probabilities used by the larger model to predict those hard labels

In the 2015 paper regarding distillation, the following passage was written:

The transfer set that is used to train the small model could consist entirely of unlabeled data [1] or we could use the original training set. We have found that using the original training set works well, especially if we add a small term to the objective function that encourages the small model to predict the true targets as well as matching the soft targets provided by the cumbersome model. Typically, the small model cannot exactly match the soft targets and erring in the direction of the correct answer turns out to be helpful.

Distilling the Knowledge in a Neural Network, Geoffery Hinton et al

Which basically meant is that we can use unlabeled data and then train the smaller model to mimic the output of the larger model. but it is preferred to use the same data the larger model was trained upon.

Now there was a slight drawback to the knowledge distillation method proposed by Hinton et al. There is a slight decrease in the accuracy when we compare the inference model and the training model. this was because it only considered the final layers of these models. To overcome this problem [Romero et al](#) proposed a solution to pick only the middle layer also called a hint layer from the inference model and transfer the knowledge to the middle layer of the inference model.

Sources: these are some sources that i found that will prove to be helpful in order to further understand distillation for our use case:

[Implementation of distilling knowledge \(not done with romero\)\)](#)

[Basic theory and understanding for Knowledge distillation](#)

[Distilbert implementation](#)

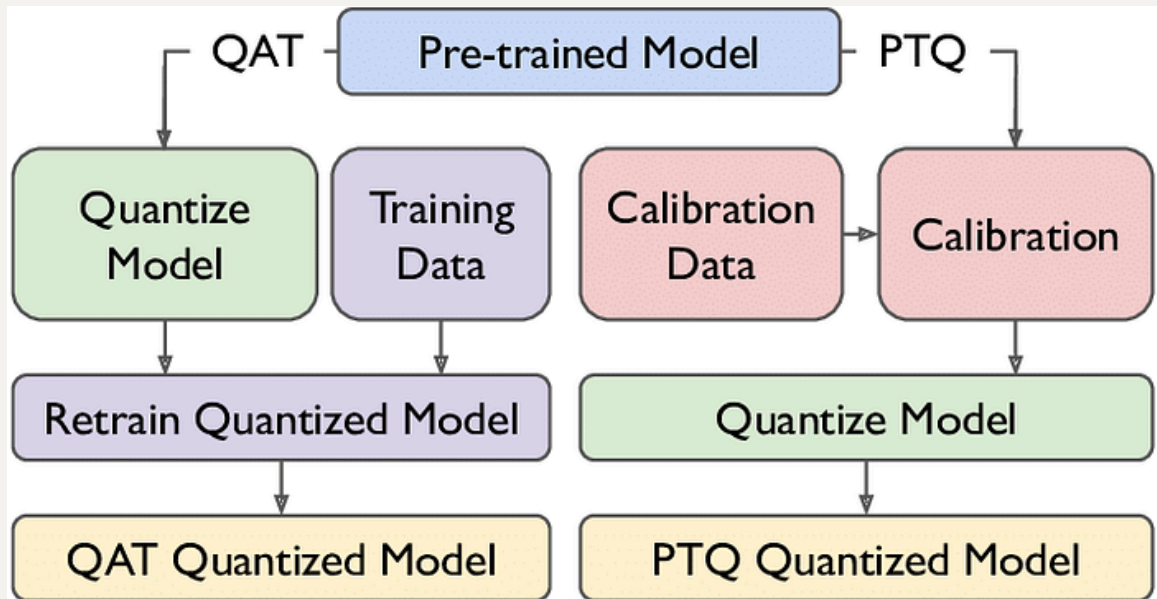
[Methods](#)

Quantization:

The main task of model quantization in deep learning is to convert high-precision floating-point numbers in neural networks into low-precision numbers.

there are two ways to perform quantization.

- **PTQ(post training quantization)**
- **QAT(quantization aware training)**



QAT(quantized aware training):

where the weights and activation functions are quantized during training of the large model.

I do not believe that QAT is relevant to our use case, therefore i will not explain it

PTQ:

a pre-trained 32bit floating point model is taken for quantization. There is no such need for training data. 500 - 2000 rows of data will be enough for PTQ to calibrate the model on.

Source:

[quantization vs binarization](#)

For our use case:

We can absolutely apply distillation for the model [serbog](#) because the data used to train this model is publicly available [here](#). Though the data is in German. so might have to use a multilingual model that is capable of understanding both German and English. The same may also be said for [JobBert V2](#) (no need for using v3, most of the data is from USA anyways.). I also think the data is somewhat sufficient: [Data](#).

We can also apply quantization on both of our models.

In fact, I believe that we should first try to quantize(PTQ) our models and see if they meet our requirements. If the accuracy is too low, then we will try to distill the models.

From looking at the hugging face repository, i think we have all of the training data for serbog, but it looks like we only have the validation data for job bert. Which i don't believe will be a problem.

We can use the following libraries to quantize and distill the models:

<https://github.com/ModelCloud/GPTQModel>

<https://paddlenlp.readthedocs.io/en/latest/llm/docs/quantization.html>

<https://arxiv.org/pdf/2410.17215> (this is a new method for distillation, and it had a good review on hugging face. I believe we will have to implement it from scratch though)

<https://github.com/microsoft/LMOps/tree/main/minillm>